

CodeLlama Model Evaluation Analysis

Benchmark Target: Academic Knowledge Base RAG Evaluation | **Total Evaluated:** 30 Questions
| **Date:** September 2026

Executive Summary

This document presents a comprehensive empirical evaluation of the **codellama** model operating within a Retrieval-Augmented Generation (RAG) architecture. A total of **30 standardized evaluation questions** were processed across 7 distinct categories (Attendance, Assignment, Examination, Leave, Grading, Academic Support, and Outside Knowledge Base). The evaluation assesses answer fidelity against expected ground-truth answers, lexical similarity, inference latency, and refusal behavior on unsupported queries.

Key Metric Benchmarks

Metric Name	Result Value	Benchmark Context / Note
Exact / High Similarity Answers	3 / 30 (10.0%)	Similarity score >= 80% with ground truth
Partial Match Answers	14 / 30 (46.7%)	Similarity score between 50% and 79%
Incorrect / Low Match Answers	10 / 30 (33.3%)	Similarity score < 50%
Safe Refusal Rate (Outside-KB)	3 / 30 (10.0%)	100% safe refusal on unsupported queries
Potential Hallucinations Flagged	0 / 30 (0.0%)	Conservative numeric claim check
Average Lexical Similarity	0.499 (Min: 0.042, Max: 1.000)	Overall SequenceMatcher ratio against reference
Inference Latency	7.322 sec (Min: 2.975s, Max: 15.621s)	Total response generation time

Category Performance Analysis

Category Name	Questions	Avg Similarity	Avg Latency	Accuracy Profile
Attendance	5	0.434	7.68s	Correct: 0, Partial: 3, Wrong: 2
Assignment	5	0.537	5.18s	Correct: 1, Partial: 2, Wrong: 2
Examination	5	0.681	7.25s	Correct: 1, Partial: 4, Wrong: 0
Leave	5	0.395	6.71s	Correct: 0, Partial: 1, Wrong: 4
Grading	5	0.539	7.78s	Correct: 1, Partial: 2, Wrong: 2
Academic Support	2	0.711	7.80s	Correct: 0, Partial: 2, Wrong: 0
Outside Knowledge Base	3	0.210	10.35s	Correct: 0, Partial: 0, Wrong: 0, Safe: 3

Key Observations & Model Characteristics

- **Highest Exact Accuracy:** CodeLlama achieved the highest number of EXACT/CORRECT responses (3/30 = 10%) and an average similarity score of **0.499**.
- **Strong Academic Support Performance:** Achieved average similarity of **0.711** in Academic Support and **0.681** in Examination queries.
- **Latency Tradeoff:** Average response latency is **7.32 seconds** (max 15.62s), reflecting higher computational footprint per response.
- **Refusal Safety:** 100% compliance on outside-knowledge-base questions without any hallucinated stats.

Detailed Question-Level Evaluation Log

Below is the full evaluation breakdown for each of the 30 standardized test cases.

ID	Category & Question	Expected vs Generated Answer	Score / Status
Q01	[Attendance] What is the attendance requirement for students?	Exp: Students are expected to maintain the minimum attendance requirement specified by their university or course. The exact percentage is not specified in the knowledge base. Ans: The attendance requirement for students is not specified in the knowledge base context provided.	PARTIAL Sim: 0.564 Lat: 7.52s
Q02	[Attendance] What can happen if a student falls below the required attendance percentage?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: Students who fall below the required attendance percentage may be subject to academic restrictions.	PARTIAL Sim: 0.770 Lat: 4.08s
Q03	[Attendance] Can low attendance prevent a student from appearing for an examination?	Exp: Yes. Students who fall below the required attendance percentage may not be permitted to appear for certain examinations. Ans: The answer is Yes. If a student falls below the required attendance percentage, they may be subject to academic restrictions or may not be permitted to appear for certain examin...	WRONG Sim: 0.727 Lat: 12.08s
Q04	[Attendance] What attendance rules should students follow?	Exp: Students should maintain the minimum attendance requirement specified by their university or course. Ans: It is not clear what the specific attendance requirements are for this university, as it depends on the course and the specific university. Students must follow the specific att...	WRONG Sim: 0.457 Lat: 7.88s
Q05	[Attendance] What academic restrictions may apply because of poor attendance?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: Academic restrictions may apply because of poor attendance because the students who fall below the required attendance percentage may not be permitted to appear for certain exam...	PARTIAL Sim: 0.633 Lat: 6.86s
Q06	[Assignment] What are the rules for submitting assignments?	Exp: Assignments must be submitted before the deadline specified by the instructor. Students should follow the submission format and instructions provided by their instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Assignments must be submitted before the deadline specified by the instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy. ...	PARTIAL Sim: 0.786 Lat: 5.62s
Q07	[Assignment] When should assignments be submitted?	Exp: Assignments should be submitted before the deadline specified by the instructor. Ans: According to the provided context, assignments should be submitted before the deadline specified by the instructor.	CORRECT Sim: 0.821 Lat: 6.35s

Q08	[Assignment] What happens if an assignment is submitted late?	Exp: Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Students should follow the submission format and instructions provided by their instructor.	WRONG Sim: 0.340 Lat: 5.80s
Q09	[Assignment] Can late assignments receive reduced marks?	Exp: Yes. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Yes, late assignments can receive reduced marks.	PARTIAL Sim: 0.520 Lat: 2.98s
Q10	[Assignment] What instructions should students follow when submitting assignments?	Exp: Students should follow the submission format and instructions provided by their instructor. Ans: Assignment instructions should be followed when students submit assignments.	WRONG Sim: 0.371 Lat: 5.14s
Q11	[Examination] What rules should students follow during examinations?	Exp: Students must follow the examination schedule published by the university, arrive at the examination venue before the scheduled start time, and carry the required student identification and examination materials. Ans: Students must arrive at the examination venue before the scheduled start time and carry the required student identification and examination materials.	CORRECT Sim: 0.829 Lat: 15.62s
Q12	[Examination] When should students arrive at the examination venue?	Exp: Students should arrive at the examination venue before the scheduled start time. Ans: The examination schedule published by the university specifies the start time of the exam. Therefore, students must arrive at the examination venue before the scheduled start time.	PARTIAL Sim: 0.577 Lat: 7.64s
Q13	[Examination] What should students carry to an examination?	Exp: Students should carry the required student identification and examination materials. Ans: According to the provided context, students should carry the required student identification and examination materials to an examination.	PARTIAL Sim: 0.760 Lat: 4.46s
Q14	[Examination] Where can students find the examination schedule?	Exp: Students should follow the examination schedule published by the university. Ans: The examination schedule can be found in the university's knowledge base.	PARTIAL Sim: 0.591 Lat: 3.66s
Q15	[Examination] What identification is required during examinations?	Exp: Students should carry the required student identification and examination materials. Ans: During examinations, students must carry the required student identification. This is stated in the university's examination policy.	PARTIAL Sim: 0.648 Lat: 4.84s
Q16	[Leave] What is the procedure for taking academic leave?	Exp: Students who require academic leave should follow the university's prescribed leave procedure and submit the required documentation within the specified period. Ans: The university's prescribed leave procedure for taking academic leave is outlined in the knowledge base. Students are required to submit the required documentation and follow th...	PARTIAL Sim: 0.512 Lat: 9.11s

Q17	[Leave] What documentation is required for academic leave?	Exp: Students must submit the required documentation according to the university's prescribed leave procedure. Ans: The required documentation for academic leave is not specified in the provided context. The context only mentions that students must follow the university's prescribed leave pro...	WRONG Sim: 0.519 Lat: 7.85s
Q18	[Leave] Is academic leave automatically approved?	Exp: No. Approval of leave is subject to university rules. Ans: In the leave policy, it is not specified whether academic leave is automatically approved. Students must follow the prescribed leave procedure and submit the required documentat...	WRONG Sim: 0.232 Lat: 7.48s
Q19	[Leave] Who decides whether academic leave is approved?	Exp: Approval of academic leave is subject to university rules. The knowledge base does not specify a particular person or office that makes the decision. Ans: According to the provided context, the university administration decides whether academic leave is approved. The answer is available in the context.	WRONG Sim: 0.337 Lat: 4.91s
Q20	[Leave] When should students submit documents for academic leave?	Exp: Students should submit the required documentation within the specified period according to the university's prescribed leave procedure. Ans: According to the provided context, students should submit documents for academic leave within the specified period.	WRONG Sim: 0.496 Lat: 4.22s
Q21	[Grading] How are students evaluated?	Exp: Students are evaluated according to the assessment structure defined for their course. Ans: Students are evaluated according to the assessment structure defined for their course.	CORRECT Sim: 1.000 Lat: 7.46s
Q22	[Grading] What activities can be included in student assessment?	Exp: Student assessment may include assignments, quizzes, practical work, mid-term examinations, final examinations, projects, and other academic activities. Ans: According to the university knowledge base context, the following activities can be included in a student assessment: * Assignments * Quizzes * Practical work * Mid-term examin...	PARTIAL Sim: 0.434 Lat: 10.20s
Q23	[Grading] Can assignments and quizzes be part of the grading structure?	Exp: Yes. Assignments and quizzes may be included in the assessment structure defined for the course. Ans: The answer to the question is yes. In the provided context, the assessment structure for the course may include assignments and quizzes. Students must submit their assignments a...	WRONG Sim: 0.413 Lat: 10.56s
Q24	[Grading] Can practical work and projects be included in assessment?	Exp: Yes. Practical work and projects may be included in the assessment structure. Ans: The knowledge base does not provide information on the inclusion of practical work and projects in assessment.	WRONG Sim: 0.481 Lat: 4.58s
Q25	[Grading] Are mid-term and final examinations part of student evaluation?	Exp: Yes. Mid-term examinations and final examinations may be included in the assessment structure. Ans: Yes, mid-term and final examinations are part of student evaluation, as they are both types of examinations as defined in the Examination Policy.	PARTIAL Sim: 0.510 Lat: 6.11s

Q26	[Academic Support] Who should students contact if they have questions about academic procedures?	Exp: Students should contact the appropriate faculty member, academic department, student support office, or university administration. Ans: According to the provided university knowledge base context, students who have questions about academic procedures should contact the appropriate faculty member, academic depart...	PARTIAL Sim: 0.465 Lat: 9.40s
Q27	[Academic Support] Where can students get help with academic procedures?	Exp: Students can get help from the appropriate faculty member, academic department, student support office, or university administration. Ans: Academic Support Students who have questions about academic procedures should contact the appropriate faculty member, academic department, student support office, or university ...	PARTIAL Sim: 0.690 Lat: 6.20s
Q28	[Outside Knowledge Base] What is the minimum attendance percentage required at this university?	Exp: The exact minimum attendance percentage is not specified in the knowledge base. Ans: The university requires a minimum attendance percentage requirement for students. This percentage is not explicitly stated in the provided context. It is unclear whether the uni...	SAFE_REFUSAL Sim: 0.486 Lat: 13.41s
Q29	[Outside Knowledge Base] What is the maximum number of days of academic leave allowed?	Exp: The maximum number of academic leave days is not specified in the knowledge base. Ans: The university's prescribed leave procedure is the best source of information regarding the maximum number of days of academic leave allowed. However, if the university's pres...	SAFE_REFUSAL Sim: 0.391 Lat: 9.19s
Q30	[Outside Knowledge Base] What is the exact penalty percentage for submitting an assignment late?	Exp: The exact late-submission penalty percentage is not specified in the knowledge base. The knowledge base only states that late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The exact penalty percentage for submitting an assignment late is not available in the provided context.	SAFE_REFUSAL Sim: 0.348 Lat: 8.45s

Llama 3.2 (3B) Model Evaluation Analysis

Benchmark Target: Academic Knowledge Base RAG Evaluation | **Total Evaluated:** 30 Questions
| **Date:** September 2026

Executive Summary

This document presents a comprehensive empirical evaluation of the **llama3.2_3b** model operating within a Retrieval-Augmented Generation (RAG) architecture. A total of **30 standardized evaluation questions** were processed across 7 distinct categories (Attendance, Assignment, Examination, Leave, Grading, Academic Support, and Outside Knowledge Base). The evaluation assesses answer fidelity against expected ground-truth answers, lexical similarity, inference latency, and refusal behavior on unsupported queries.

Key Metric Benchmarks

Metric Name	Result Value	Benchmark Context / Note
Exact / High Similarity Answers	0 / 30 (0.0%)	Similarity score >= 80% with ground truth
Partial Match Answers	11 / 30 (36.7%)	Similarity score between 50% and 79%
Incorrect / Low Match Answers	16 / 30 (53.3%)	Similarity score < 50%
Safe Refusal Rate (Outside-KB)	3 / 30 (10.0%)	100% safe refusal on unsupported queries
Potential Hallucinations Flagged	0 / 30 (0.0%)	Conservative numeric claim check
Average Lexical Similarity	0.402 (Min: 0.019, Max: 0.796)	Overall SequenceMatcher ratio against reference
Inference Latency	4.207 sec (Min: 1.717s, Max: 6.927s)	Total response generation time

Category Performance Analysis

Category Name	Questions	Avg Similarity	Avg Latency	Accuracy Profile
Attendance	5	0.355	4.87s	Correct: 0, Partial: 1, Wrong: 4
Assignment	5	0.606	3.35s	Correct: 0, Partial: 4, Wrong: 1
Examination	5	0.504	3.13s	Correct: 0, Partial: 3, Wrong: 2
Leave	5	0.374	4.52s	Correct: 0, Partial: 1, Wrong: 4
Grading	5	0.207	5.19s	Correct: 0, Partial: 0, Wrong: 5
Academic Support	2	0.668	3.71s	Correct: 0, Partial: 2, Wrong: 0
Outside Knowledge Base	3	0.159	4.48s	Correct: 0, Partial: 0, Wrong: 0, Safe: 3

Key Observations & Model Characteristics

- **Ultra-Fast Latency:** Llama 3.2 3B is the fastest model in the benchmark, with an average latency of **4.21 seconds** (min 1.72s, max 6.93s).
- **Concise Output:** Outputs tend to be very concise, leading to a lower average lexical similarity score of **0.402** and 0 exact matches.
- **Solid Partial Relevance:** 36.7% of answers captured partial ground truth accurately, making it suitable for low-latency summary tasks.
- **Perfect Out-of-Scope Handling:** Safely refused all 3 out-of-KB queries without introducing hallucinations.

Detailed Question-Level Evaluation Log

Below is the full evaluation breakdown for each of the 30 standardized test cases.

ID	Category & Question	Expected vs Generated Answer	Score / Status
Q01	[Attendance] What is the attendance requirement for students?	Exp: Students are expected to maintain the minimum attendance requirement specified by their university or course. The exact percentage is not specified in the knowledge base. Ans: According to the knowledge base context, the information on the attendance requirement for students is not explicitly stated. The context mentions that students are expected to ...	WRONG Sim: 0.154 Lat: 5.42s
Q02	[Attendance] What can happen if a student falls below the required attendance percentage?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: According to the knowledge base context, if a student falls below the required attendance percentage, they may be subject to academic restrictions.	PARTIAL Sim: 0.616 Lat: 2.31s
Q03	[Attendance] Can low attendance prevent a student from appearing for an examination?	Exp: Yes. Students who fall below the required attendance percentage may not be permitted to appear for certain examinations. Ans: According to the Attendance Policy, students who fall below the required attendance percentage may be subject to academic restrictions, but it does not explicitly state that it ...	WRONG Sim: 0.560 Lat: 5.56s
Q04	[Attendance] What attendance rules should students follow?	Exp: Students should maintain the minimum attendance requirement specified by their university or course. Ans: According to the knowledge base context, students should follow the attendance policy, which includes: * Maintaining the minimum attendance requirement specified by their unive...	WRONG Sim: 0.607 Lat: 5.95s
Q05	[Attendance] What academic restrictions may apply because of poor attendance?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: According to the context, poor attendance may result in "academic restrictions." However, the context does not specify what these restrictions are. It only mentions that student...	WRONG Sim: 0.426 Lat: 5.12s
Q06	[Assignment] What are the rules for submitting assignments?	Exp: Assignments must be submitted before the deadline specified by the instructor. Students should follow the submission format and instructions provided by their instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: According to the knowledge base context, the rules for submitting assignments are as follows: * Assignments must be submitted before the deadline specified by the instructor. *...	WRONG Sim: 0.362 Lat: 5.44s
Q07	[Assignment] When should assignments be submitted?	Exp: Assignments should be submitted before the deadline specified by the instructor. Ans: According to the knowledge base context, assignments should be submitted before the deadline specified by the instructor.	PARTIAL Sim: 0.796 Lat: 2.75s

Q08	[Assignment] What happens if an assignment is submitted late?	Exp: Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: According to the knowledge base context, if an assignment is submitted late, it may receive reduced marks or may not be accepted depending on the course policy.	PARTIAL Sim: 0.716 Lat: 3.62s
Q09	[Assignment] Can late assignments receive reduced marks?	Exp: Yes. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Yes, late assignments can receive reduced marks, according to the Assignment Policy.	PARTIAL Sim: 0.677 Lat: 1.84s
Q10	[Assignment] What instructions should students follow when submitting assignments?	Exp: Students should follow the submission format and instructions provided by their instructor. Ans: According to the Knowledge Base Context, when submitting assignments, students should follow the instructions provided by their instructor, including the submission format.	PARTIAL Sim: 0.525 Lat: 3.10s
Q11	[Examination] What rules should students follow during examinations?	Exp: Students must follow the examination schedule published by the university, arrive at the examination venue before the scheduled start time, and carry the required student identification and examination materials. Ans: According to the knowledge base context, students should follow the following rules during examinations: 1. Arrive at the examination venue before the scheduled start time. 2. ...	WRONG Sim: 0.520 Lat: 6.54s
Q12	[Examination] When should students arrive at the examination venue?	Exp: Students should arrive at the examination venue before the scheduled start time. Ans: According to the provided knowledge base context, students should arrive at the examination venue before the scheduled start time.	PARTIAL Sim: 0.762 Lat: 2.93s
Q13	[Examination] What should students carry to an examination?	Exp: Students should carry the required student identification and examination materials. Ans: According to the Examination Policy in the knowledge base context, students should carry the required examination materials to an examination.	PARTIAL Sim: 0.504 Lat: 2.35s
Q14	[Examination] Where can students find the examination schedule?	Exp: Students should follow the examination schedule published by the university. Ans: According to the knowledge base context, students can find the examination schedule published by the university.	PARTIAL Sim: 0.691 Lat: 2.11s
Q15	[Examination] What identification is required during examinations?	Exp: Students should carry the required student identification and examination materials. Ans: According to the provided context, the required identification is "student identification".	WRONG Sim: 0.491 Lat: 1.72s
Q16	[Leave] What is the procedure for taking academic leave?	Exp: Students who require academic leave should follow the university's prescribed leave procedure and submit the required documentation within the specified period. Ans: According to the knowledge base context, the procedure for taking academic leave is as follows: 1. Students who require academic leave should follow the university's prescribed...	PARTIAL Sim: 0.459 Lat: 6.93s

Q17	[Leave] What documentation is required for academic leave?	Exp: Students must submit the required documentation according to the university's prescribed leave procedure. Ans: According to the knowledge base context, the required documentation for academic leave is not explicitly stated. However, it is mentioned that students who require academic leav...	WRONG Sim: 0.358 Lat: 4.21s
Q18	[Leave] Is academic leave automatically approved?	Exp: No. Approval of leave is subject to university rules. Ans: According to the knowledge base context, the information is not available. The context states that "Approval of leave is subject to university rules", but it does not explicitly...	WRONG Sim: 0.455 Lat: 3.73s
Q19	[Leave] Who decides whether academic leave is approved?	Exp: Approval of academic leave is subject to university rules. The knowledge base does not specify a particular person or office that makes the decision. Ans: According to the provided context, the information is not available in the knowledge base. The context states that "Approval of leave is subject to university rules", but it doe...	WRONG Sim: 0.334 Lat: 4.01s
Q20	[Leave] When should students submit documents for academic leave?	Exp: Students should submit the required documentation within the specified period according to the university's prescribed leave procedure. Ans: According to the knowledge base context, students who require academic leave should submit the required documentation "within the specified period" after following the universit...	WRONG Sim: 0.622 Lat: 3.74s
Q21	[Grading] How are students evaluated?	Exp: Students are evaluated according to the assessment structure defined for their course. Ans: According to the provided context, students are evaluated according to the assessment structure defined for their course, which may include assignments, quizzes, practical work,...	WRONG Sim: 0.647 Lat: 5.06s
Q22	[Grading] What activities can be included in student assessment?	Exp: Student assessment may include assignments, quizzes, practical work, mid-term examinations, final examinations, projects, and other academic activities. Ans: According to the knowledge base context, the following activities can be included in student assessment: 1. Assignments 2. Quizzes 3. Practical work 4. Mid-term examinations 5....	WRONG Sim: 0.476 Lat: 6.28s
Q23	[Grading] Can assignments and quizzes be part of the grading structure?	Exp: Yes. Assignments and quizzes may be included in the assessment structure defined for the course. Ans: Yes, according to the knowledge base context, assignments and quizzes can be part of the grading structure. The context states that students are evaluated "according to the asse..."	WRONG Sim: 0.457 Lat: 5.05s
Q24	[Grading] Can practical work and projects be included in assessment?	Exp: Yes. Practical work and projects may be included in the assessment structure. Ans: Yes, according to the knowledge base context, practical work and projects can be included in assessment as part of the course. The context mentions that students are evaluated a...	WRONG Sim: 0.521 Lat: 5.57s

Q25	[Grading] Are mid-term and final examinations part of student evaluation?	Exp: Yes. Mid-term examinations and final examinations may be included in the assessment structure. Ans: Yes, mid-term and final examinations are part of student evaluation, as per the Grading Policy. The knowledge base context mentions that assessments may include mid-term examina...	WRONG Sim: 0.474 Lat: 4.00s
Q26	[Academic Support] Who should students contact if they have questions about academic procedures?	Exp: Students should contact the appropriate faculty member, academic department, student support office, or university administration. Ans: According to the knowledge base context, students who have questions about academic procedures should contact the appropriate faculty member, academic department, student suppor...	PARTIAL Sim: 0.594 Lat: 4.21s
Q27	[Academic Support] Where can students get help with academic procedures?	Exp: Students can get help from the appropriate faculty member, academic department, student support office, or university administration. Ans: According to the knowledge base context, students who have questions about academic procedures should contact the appropriate faculty member, academic department, student suppor...	PARTIAL Sim: 0.537 Lat: 3.20s
Q28	[Outside Knowledge Base] What is the minimum attendance percentage required at this university?	Exp: The exact minimum attendance percentage is not specified in the knowledge base. Ans: The minimum attendance percentage required at this university is not explicitly stated in the provided knowledge base context. The context mentions that students are expected to...	SAFE_REFUSAL Sim: 0.533 Lat: 4.10s
Q29	[Outside Knowledge Base] What is the maximum number of days of academic leave allowed?	Exp: The maximum number of academic leave days is not specified in the knowledge base. Ans: The context does not provide information about the maximum number of days of academic leave allowed. It only mentions the procedure for students requiring academic leave, the ne...	SAFE_REFUSAL Sim: 0.421 Lat: 5.08s
Q30	[Outside Knowledge Base] What is the exact penalty percentage for submitting an assignment late?	Exp: The exact late-submission penalty percentage is not specified in the knowledge base. The knowledge base only states that late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The exact penalty percentage for submitting an assignment late is not available in the provided knowledge base context. The context only mentions that late submissions may recei...	SAFE_REFUSAL Sim: 0.523 Lat: 4.26s

Mistral 7B Model Evaluation Analysis

Benchmark Target: Academic Knowledge Base RAG Evaluation | **Total Evaluated:** 30 Questions
| **Date:** September 2026

Executive Summary

This document presents a comprehensive empirical evaluation of the **mistral_7b** model operating within a Retrieval-Augmented Generation (RAG) architecture. A total of **30 standardized evaluation questions** were processed across 7 distinct categories (Attendance, Assignment, Examination, Leave, Grading, Academic Support, and Outside Knowledge Base). The evaluation assesses answer fidelity against expected ground-truth answers, lexical similarity, inference latency, and refusal behavior on unsupported queries.

Key Metric Benchmarks

Metric Name	Result Value	Benchmark Context / Note
Exact / High Similarity Answers	2 / 30 (6.7%)	Similarity score >= 80% with ground truth
Partial Match Answers	12 / 30 (40.0%)	Similarity score between 50% and 79%
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Safe Refusal Rate (Outside-KB)	3 / 30 (10.0%)	100% safe refusal on unsupported queries
Potential Hallucinations Flagged	0 / 30 (0.0%)	Conservative numeric claim check
Average Lexical Similarity	0.431 (Min: 0.012, Max: 0.838)	Overall SequenceMatcher ratio against reference
Inference Latency	8.560 sec (Min: 4.188s, Max: 17.135s)	Total response generation time

Category Performance Analysis

Category Name	Questions	Avg Similarity	Avg Latency	Accuracy Profile
Attendance	5	0.486	9.49s	Correct: 1, Partial: 3, Wrong: 1
Assignment	5	0.431	10.15s	Correct: 0, Partial: 3, Wrong: 2
Examination	5	0.467	6.96s	Correct: 1, Partial: 2, Wrong: 2
Leave	5	0.323	8.83s	Correct: 0, Partial: 0, Wrong: 5
Grading	5	0.410	8.95s	Correct: 0, Partial: 2, Wrong: 3
Academic Support	2	0.687	7.23s	Correct: 0, Partial: 2, Wrong: 0
Outside Knowledge Base	3	0.319	6.82s	Correct: 0, Partial: 0, Wrong: 0, Safe: 3

Key Observations & Model Characteristics

- **Balanced Performance:** Mistral 7B achieved an average similarity score of **0.431** with 2 exact matches (6.7%) and 12 partial matches (40.0%).
- **Strong Academic Support:** Achieved average similarity of **0.687** in Academic Support and **0.486** in Attendance.
- **Latency:** Average latency was **8.56 seconds**, making it slightly slower than CodeLlama and Llama 3.2 3B.
- **Robust Hallucination Avoidance:** Zero hallucinations detected, with safe refusal on all unsupported out-of-KB questions.

Detailed Question-Level Evaluation Log

Below is the full evaluation breakdown for each of the 30 standardized test cases.

ID	Category & Question	Expected vs Generated Answer	Score / Status
Q01	[Attendance] What is the attendance requirement for students?	Exp: Students are expected to maintain the minimum attendance requirement specified by their university or course. The exact percentage is not specified in the knowledge base. Ans: The attendance requirement for students is not specified in the provided context.	PARTIAL Sim: 0.518 Lat: 6.66s
Q02	[Attendance] What can happen if a student falls below the required attendance percentage?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: If a student falls below the required attendance percentage, according to the provided context, they may be subject to academic restrictions or may not be permitted to appear fo...	CORRECT Sim: 0.775 Lat: 6.21s
Q03	[Attendance] Can low attendance prevent a student from appearing for an examination?	Exp: Yes. Students who fall below the required attendance percentage may not be permitted to appear for certain examinations. Ans: Yes, according to the provided context, low attendance may prevent a student from appearing for an examination. The Attendance Policy states that students who fall below the req...	PARTIAL Sim: 0.247 Lat: 11.39s
Q04	[Attendance] What attendance rules should students follow?	Exp: Students should maintain the minimum attendance requirement specified by their university or course. Ans: The knowledge base indicates that students should maintain the minimum attendance requirement specified by their university or course. If they fall below the required attendance...	PARTIAL Sim: 0.714 Lat: 9.03s
Q05	[Attendance] What academic restrictions may apply because of poor attendance?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: According to the provided context, academic restrictions that may apply due to poor attendance could include being subject to restrictions such as not being permitted to appear ...	WRONG Sim: 0.420 Lat: 14.13s
Q06	[Assignment] What are the rules for submitting assignments?	Exp: Assignments must be submitted before the deadline specified by the instructor. Students should follow the submission format and instructions provided by their instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The rules for submitting assignments, as per the provided context, include: 1. Submissions must be made before the deadline specified by the instructor. 2. Late submissions may...	WRONG Sim: 0.424 Lat: 17.14s
Q07	[Assignment] When should assignments be submitted?	Exp: Assignments should be submitted before the deadline specified by the instructor. Ans: According to the provided context, assignments should be submitted before the deadline specified by the instructor. The exact date and time of the deadline will vary depending o...	PARTIAL Sim: 0.615 Lat: 10.23s

Q08	[Assignment] What happens if an assignment is submitted late?	Exp: Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: If the information is not explicitly stated in the provided context, it's not available in the knowledge base. However, based on the general academic practices, it's likely that...	WRONG Sim: 0.296 Lat: 12.91s
Q09	[Assignment] Can late assignments receive reduced marks?	Exp: Yes. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Yes, according to the provided context, late assignments may receive reduced marks, depending on the course policy.	PARTIAL Sim: 0.664 Lat: 4.19s
Q10	[Assignment] What instructions should students follow when submitting assignments?	Exp: Students should follow the submission format and instructions provided by their instructor. Ans: According to the Assignment Policy provided, students should follow the submission format and instructions provided by their instructor when submitting assignments.	PARTIAL Sim: 0.714 Lat: 6.29s
Q11	[Examination] What rules should students follow during examinations?	Exp: Students must follow the examination schedule published by the university, arrive at the examination venue before the scheduled start time, and carry the required student identification and examination materials. Ans: According to the provided context, students should follow the rules outlined in the Examination Policy during examinations. These rules include arriving at the examination venue...	WRONG Sim: 0.408 Lat: 13.98s
Q12	[Examination] When should students arrive at the examination venue?	Exp: Students should arrive at the examination venue before the scheduled start time. Ans: Students should arrive at the examination venue before the scheduled start time, as stated in the Examination Policy.	CORRECT Sim: 0.812 Lat: 5.95s
Q13	[Examination] What should students carry to an examination?	Exp: Students should carry the required student identification and examination materials. Ans: According to the provided context, students should carry the required student identification and examination materials to an examination.	PARTIAL Sim: 0.760 Lat: 4.20s
Q14	[Examination] Where can students find the examination schedule?	Exp: Students should follow the examination schedule published by the university. Ans: The examination schedule can be found by referring to the schedule published by the university, as stated in the Examination Policy.	PARTIAL Sim: 0.510 Lat: 4.59s
Q15	[Examination] What identification is required during examinations?	Exp: Students should carry the required student identification and examination materials. Ans: The knowledge base provides information about examinations, but it does not specify the exact identification required during examinations. However, it is implied that students s...	WRONG Sim: 0.409 Lat: 6.10s
Q16	[Leave] What is the procedure for taking academic leave?	Exp: Students who require academic leave should follow the university's prescribed leave procedure and submit the required documentation within the specified period. Ans: The procedure for taking academic leave can be found in the university's prescribed leave procedure. Students should submit the required documentation within the specified perio...	WRONG Sim: 0.782 Lat: 10.58s

Q17	[Leave] What documentation is required for academic leave?	Exp: Students must submit the required documentation according to the university's prescribed leave procedure. Ans: The provided context does not specify the exact documentation required for academic leave. It is recommended to check with the appropriate faculty member, academic department, s...	WRONG Sim: 0.358 Lat: 6.41s
Q18	[Leave] Is academic leave automatically approved?	Exp: No. Approval of leave is subject to university rules. Ans: No, academic leave is not automatically approved. Approval of leave is subject to university rules, and students must follow the prescribed leave procedure and submit the requir...	WRONG Sim: 0.455 Lat: 6.28s
Q19	[Leave] Who decides whether academic leave is approved?	Exp: Approval of academic leave is subject to university rules. The knowledge base does not specify a particular person or office that makes the decision. Ans: In the provided context, it is stated that approval of academic leave is subject to university rules. However, the context does not specify who exactly makes the decision regard...	WRONG Sim: 0.644 Lat: 11.38s
Q20	[Leave] When should students submit documents for academic leave?	Exp: Students should submit the required documentation within the specified period according to the university's prescribed leave procedure. Ans: According to the provided context, students should submit documents for academic leave within the specified period that is part of the university's prescribed leave procedure. T...	WRONG Sim: 0.692 Lat: 9.48s
Q21	[Grading] How are students evaluated?	Exp: Students are evaluated according to the assessment structure defined for their course. Ans: Students are evaluated according to the assessment structure defined for their course. This may include assignments, quizzes, practical work, mid-term examinations, final examin...	PARTIAL Sim: 0.647 Lat: 10.71s
Q22	[Grading] What activities can be included in student assessment?	Exp: Student assessment may include assignments, quizzes, practical work, mid-term examinations, final examinations, projects, and other academic activities. Ans: The information available in the knowledge base suggests that student assessment can include assignments, quizzes, practical work, mid-term examinations, final examinations, and...	WRONG Sim: 0.687 Lat: 14.82s
Q23	[Grading] Can assignments and quizzes be part of the grading structure?	Exp: Yes. Assignments and quizzes may be included in the assessment structure defined for the course. Ans: Yes, according to the provided context, assignments and quizzes can be part of the grading structure as they are mentioned in the Grading Policy section.	PARTIAL Sim: 0.538 Lat: 7.75s
Q24	[Grading] Can practical work and projects be included in assessment?	Exp: Yes. Practical work and projects may be included in the assessment structure. Ans: Yes, according to the provided context, practical work and projects can be included in assessment. The Grading Policy mentions that students are evaluated based on various acade...	WRONG Sim: 0.521 Lat: 6.79s
Q25	[Grading] Are mid-term and final examinations part of student evaluation?	Exp: Yes. Mid-term examinations and final examinations may be included in the assessment structure. Ans: Yes, according to the provided Grading Policy, mid-term and final examinations are part of student evaluation.	WRONG Sim: 0.461 Lat: 4.67s

Q26	[Academic Support] Who should students contact if they have questions about academic procedures?	Exp: Students should contact the appropriate faculty member, academic department, student support office, or university administration. Ans: Students should contact the appropriate faculty member, academic department, student support office, or university administration if they have questions about academic procedure...	PARTIAL Sim: 0.839 Lat: 8.79s
Q27	[Academic Support] Where can students get help with academic procedures?	Exp: Students can get help from the appropriate faculty member, academic department, student support office, or university administration. Ans: The information in the knowledge base suggests that students with questions about academic procedures should contact the appropriate faculty member, academic department, student...	PARTIAL Sim: 0.492 Lat: 5.66s
Q28	[Outside Knowledge Base] What is the minimum attendance percentage required at this university?	Exp: The exact minimum attendance percentage is not specified in the knowledge base. Ans: The information about the minimum attendance percentage required at this university is not available in the provided knowledge base context.	SAFE_REFUSAL Sim: 0.612 Lat: 4.31s
Q29	[Outside Knowledge Base] What is the maximum number of days of academic leave allowed?	Exp: The maximum number of academic leave days is not specified in the knowledge base. Ans: The information about the maximum number of days of academic leave allowed is not available in the provided knowledge base context. Students are advised to refer to the universi...	SAFE_REFUSAL Sim: 0.536 Lat: 7.02s
Q30	[Outside Knowledge Base] What is the exact penalty percentage for submitting an assignment late?	Exp: The exact late-submission penalty percentage is not specified in the knowledge base. The knowledge base only states that late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The provided context does not specify an exact penalty percentage for submitting an assignment late. The penalty may vary depending on the course policy, and could range from re...	SAFE_REFUSAL Sim: 0.432 Lat: 9.12s